

# Montrac OPC UA Interface Manual

Description of OPC UA methods and their parameters.

---

W1/W2 Montrac

V2

2026-01-07

Jiří Štefan

jiri.stefan@cvut.cz



## CONTENTS

|                                     |   |
|-------------------------------------|---|
| Interface Overview .....            | 2 |
| Method: Start Robot Operation ..... | 3 |
| Method: Toggle Robot .....          | 5 |
| Method: Send Shuttle .....          | 6 |
| Method: Toggle Station Lock .....   | 7 |
| Call Status Codes .....             | 8 |

## INTERFACE OVERVIEW

All important nodes are under the **W1** node in the information tree.

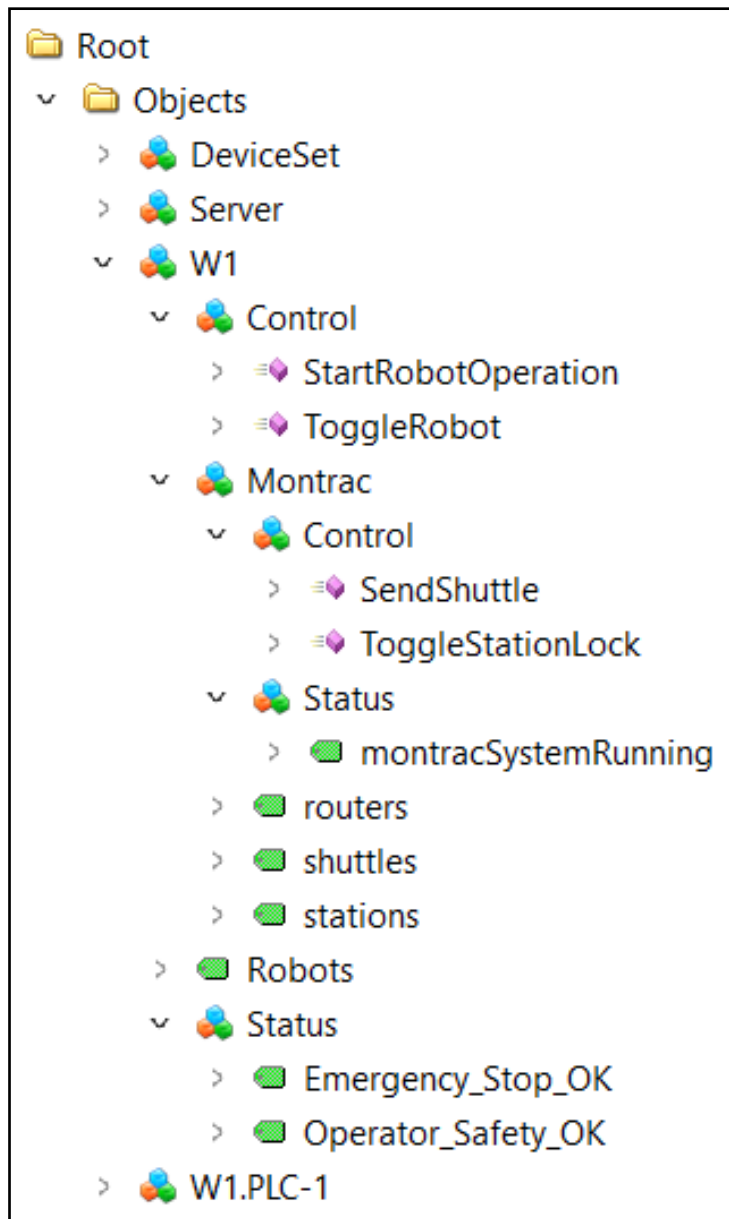


Figure 1: Montrac OPC UA interface tree.

## METHOD: START ROBOT OPERATION

Starts selected operation on a robot.

The screenshot shows a dialog box titled "Call StartRobotOperation on Control". It has two main sections: "Input Arguments" and "Output Arguments".

**Input Arguments:**

| Name             | Value                             | DataType | Description |
|------------------|-----------------------------------|----------|-------------|
| Robot_ID         | 1                                 | Byte     |             |
| Communication_ID | MY_ID                             | String   |             |
| Program_Number   | 50                                | UInt32   |             |
| Parameter_Array  | {0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0} | Int32    |             |

**Output Arguments:**

| Name             | Value          | DataType | Description |
|------------------|----------------|----------|-------------|
| Call_ID          | 0              | UInt32   |             |
| Communication_ID |                | String   |             |
| Result_Code      | 100            | UInt32   |             |
| Result_Message   | Emergency stop | String   |             |

**Result:** Succeeded

Figure 2: Example method call in the UaExpert software tool.

| Number | Name                       |
|--------|----------------------------|
| 10     | Move To Home               |
| 11     | Move To Position           |
| 20     | Pick                       |
| 21     | Pick Vertical              |
| 30     | Place                      |
| 31     | Place Vertical             |
| 40     | Approach Position          |
| 41     | Approach Position Vertical |
| 50     | Brake Test                 |
| 51     | Run Tool Command           |

Table 1: List of available robot operations.

Every call has required arguments that need to be filled in, as well as an optional array of generic parameters, which are handled according to the type of operation that is to be performed.

**Robot\_ID** → 8-bit Unsigned Integer

ID of the robot which should perform the operation.

**Communication\_ID** → String

User-defined communication ID which is directly returned as an output of the method call, unmodified. This is not used by the system, and only serves as an optional identifier for the method caller.

**Program\_Number** → 32-bit Unsigned Integer

Number which identifies the type of the robot operation which should be performed.

**Parameter\_Array** → 32-bit Signed Integer (20x)

Array of generic numbers, which can be used to pass additional operation-specific information to the robot.

## METHOD: TOGGLE ROBOT

Disables robot at a software level. The robot will still be powered on, but will not perform any operations, and will show different beacon light.

| Input Arguments |                          |          |             |
|-----------------|--------------------------|----------|-------------|
| Name            | Value                    | DataType | Description |
| Robot_ID        | 1                        | Byte     |             |
| Disable         | <input type="checkbox"/> | Boolean  |             |

| Output Arguments |         |          |             |
|------------------|---------|----------|-------------|
| Name             | Value   | DataType | Description |
| Result_Code      | 0       | UInt32   |             |
| Result_Message   | Success | String   | ...         |

Result: Succeeded

Figure 3: Example method call in the UaExpert software tool.

**Robot\_ID** → 8-bit Unsigned Integer

ID of the robot which should perform the operation.

**Disable** → Bool

When TRUE, the call disables the specified robot. When FALSE, the call enables the specified robot. Enabling or disabling a robot multiple times in a row does not return an error, and does not change the state of the robot further.

## METHOD: SEND SHUTTLE

Sends a shuttle to another station.

| Input Arguments     |                                     |          |             |
|---------------------|-------------------------------------|----------|-------------|
| Name                | Value                               | DataType | Description |
| Shuttle_ID          | 3                                   | Byte     |             |
| Communication_ID    | MY_ID                               | String   |             |
| Target_Station_Name | S300                                | String   |             |
| Lock_After_Arrival  | <input checked="" type="checkbox"/> | Boolean  |             |

| Output Arguments |                    |          |             |
|------------------|--------------------|----------|-------------|
| Name             | Value              | DataType | Description |
| Communication_ID |                    | String   |             |
| Result_Code      | 2005               | UInt32   |             |
| Result_Message   | Invalid shuttle ID | String   |             |

| Result    |  |
|-----------|--|
| Succeeded |  |

Figure 4: Example method call in the UaExpert software tool.

**Shuttle\_ID** → 8-bit Unsigned Integer

ID of the robot which should perform the operation.

**Communication\_ID** → String

User-defined communication ID which is directly returned as an output of the method call, unmodified. This is not used by the system, and only serves as an optional identifier for the user.

**Target\_Station\_Name** → String

Name of the station the shuttle should travel to.

**Lock\_After\_Arrival** → Bool

When TRUE, the target station lock locks the shuttle in place when it arrives. When FALSE, the shuttle remains unlocked after arrival.

## METHOD: TOGGLE STATION LOCK

Changes the state on Montrac station lock. Only stations with a shuttle present can be locked, for safety reasons.



Figure 5: Example method call in the UaExpert software tool.

**Station\_Name** → String

Name of the station the shuttle should travel to.

**Lock** → Bool

When TRUE, the target station is locked. When FALSE, the target station is unlocked. Locking or unlocking a station multiple times in a row does not return an error, and does not change the state of the lock further.

## CALL STATUS CODES

| Code | Type    | Meaning                  |
|------|---------|--------------------------|
| 0    | Basic   | Success                  |
| 1    | Basic   | Internal error           |
| 2    | Basic   | Invalid parameters       |
| 3    | Basic   | Timeout                  |
| 100  | General | Emergency stop           |
| 101  | General | Operator safety          |
| 1000 | Robot   | ID invalid               |
| 1001 | Robot   | Not ready                |
| 1002 | Robot   | Error active             |
| 1003 | Robot   | Program start failed     |
| 1004 | Robot   | Disabled                 |
| 2000 | Montrac | Station name invalid     |
| 2001 | Montrac | Monorail not running     |
| 2002 | Montrac | Station cannot be locked |
| 2003 | Montrac | Shuttle not in station   |
| 2004 | Montrac | Station lock error       |
| 2005 | Montrac | Shuttle ID invalid       |
| 2006 | Montrac | Departure prohibited     |
| 2007 | Montrac | Shuttle failed to depart |

Table 2: List of status codes returned by OPC UA method calls.